High school · optimisation and olympiads

Inequalities and equality cases

Synthesis problem

Investigate, justify, conclude

There are 40 m of fencing for three sides of a rectangular enclosure against a wall. Its width x, perpendicular to the wall, is positive. Express its length and area in terms of x, determine the domain, then prove the maximum area without derivatives. Compare with a fully fenced rectangle of perimeter 40 m.